

Murilo Dias

Backend Developer

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Summary

Backend Developer focused on **JavaScript** and **TypeScript** within the **Node.js** ecosystem, I work on building APIs and real-time systems. A Computer Engineering undergraduate, I developed a web system for the Andradina Health Surveillance Agency that reduced complaint lookup time **from 5 minutes to 10 seconds**, replacing manual processes with an intranet solution.

Technical Skills

- **Languages:** JavaScript (ES6+), TypeScript
- **Backend:** Node.js, Express.js, REST APIs
- **Real-Time & Media:** WebSocket (Socket.io), WebRTC, FFmpeg
- **Data:** PostgreSQL, MariaDB (MySQL), MongoDB (Prisma, Sequelize)
- **Infrastructure & Tooling:** Linux, Nginx (reverse proxy), Docker, PM2, AWS (EC2), Git

Professional Experience

Andradina City Hall ([GitHub](#))
Backend Developer (Node.js)

Andradina, SP
May 2023 – Present

- Reduced audit response time **from 5 minutes to 10 seconds** by designing a REST API in **Node.js (JavaScript)** to centralize the registration and lookup of Health Surveillance complaints.
- Structured the relational database in **MariaDB**, modeling entities and relationships with **Sequelize ORM** and applying normalization and indexing to optimize queries.
- Developed the front-end with **HTML**, **CSS**, **JavaScript**, **Bootstrap**, and **EJS**, building interfaces for complaint management and report generation, improving inspectors' operational efficiency.
- Ensured continuous availability in a **Linux** environment via intranet with **PM2** and **Nginx**, restricting access to the internal network and eliminating the need for external authentication.

Andradina City Hall
Support and Systems Intern

Andradina, SP
June 2022 – April 2023

- Reduced service time by 60% by developing a WhatsApp bot in **Node.js (JavaScript)** with **whatsapp-web.js**.
- Optimized **SQL** database response time by 30% by applying **indexing** and **caching** strategies.
- Automated administrative routines using **Node.js (JavaScript)** scripts, reducing manual tasks.

Engineering Projects

Integra — Education Without Barriers ([Demo](#))
Full-Stack Developer

Capstone Project (Academic Distinction)
2025

- Developed a speech-to-Libras translation system with a 3D avatar in **Node.js (JavaScript)**, integrating and extending **VLibras** through interception of internal events (*event patching*).
- Implemented a real-time audio pipeline with **WebSocket** and **FFmpeg**, including bidirectional communication, continuous frame sending, metadata handshake, and session control.
- Integrated multiple speech recognition engines (**Vosk** and **Google Speech-to-Text**) with dynamic selection.
- Developed the front-end with **HTML**, **CSS**, **JavaScript**, **Bootstrap**, and **EJS**.

Education

UniSALESIANO Araçatuba
Bachelor's Degree in Computer Engineering

Araçatuba, SP
Expected graduation: Dec 2026

Excellent Global
Advanced English (C1)

Andradina, SP